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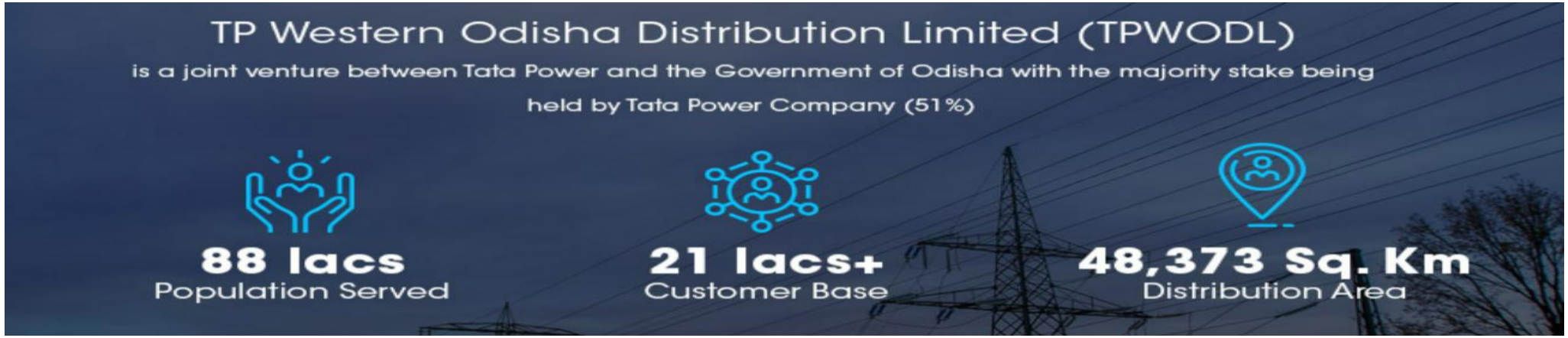
# Session : Smart Metering and Smart Meter Data Analytics

## Smart Meter Data Analytics

*Presented By*

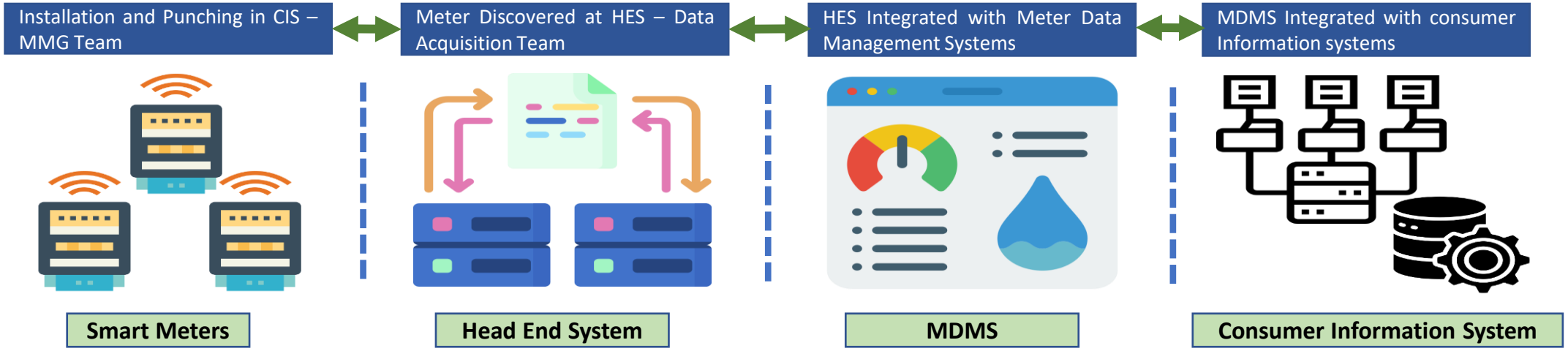
*Rupesh Goyal, Head-MMG, TPWODL*

# INTRODUCTION




- **TP Western Odisha Distribution Limited (TPWODL)** is a joint venture of Tata Power & Govt. of Odisha, responsible for electricity distribution in western Odisha since 1 January 2021.
- At takeover, the utility faced **high AT&C losses (28.56%),** low operational visibility, and process inefficiencies.
- **Billing Efficiency: 78%, Collection Efficiency: 91.6%** in **FY 2019–20** indicated significant gaps in revenue assurance.
- **TPWODL adopted a multi-layered turnaround strategy**—technology integration, asset strengthening, digitalisation, and process discipline.
- By **FY 2024–25**, AT&C losses reduced to **16.23 %** and **Collection Efficiency** improved to **99%.**

# CONTEXT - SMART METER DATA UTILISATION




- **<u>Large-scale smart metering rollout accelerated loss reduction</u>—100% coverage for LTCT & WC** consumers (**1.04 lakh meters**) and **35% penetration in single-phase consumers**.
- To **maximise value from digital data**, TPWODL created the **Meter Data Analysis Group (MDAG)—**a small team dedicated to analysing smart meter data and detecting anomalies, theft, and under-recording.
- Smart meters generate **high-frequency granular** data: **15-minute load** profiles, daily profiles, billing data, events, and alerts.
- **MDAG analyses** this data using **customised logics** and correlates it with consumer master data.
- Helps detect theft, tampering, under-recording, and meter malfunctions with high precision.
- All **suspicious cases** are forwarded to **Enforcement Teams** for field verification and corrective action.

## Detection Logics Developed by MDAG

- ✓ Load without potential
- ✓ Abnormal / high neutral current
- ✓ Tamper events: cover open, ESD, magnet influence, etc.
- ✓ Phase-wise current mismatch
- ✓ Neutral missing or improper earthing conditions
- ✓ High voltage / meter overheating alerts
- ✓ Consumption assessment using MD trends

**Logics accurately identify anomalies, leakage windows, and event time stamps, improving on-ground action and revenue protection.**

# USE CASE / CASE STUDY



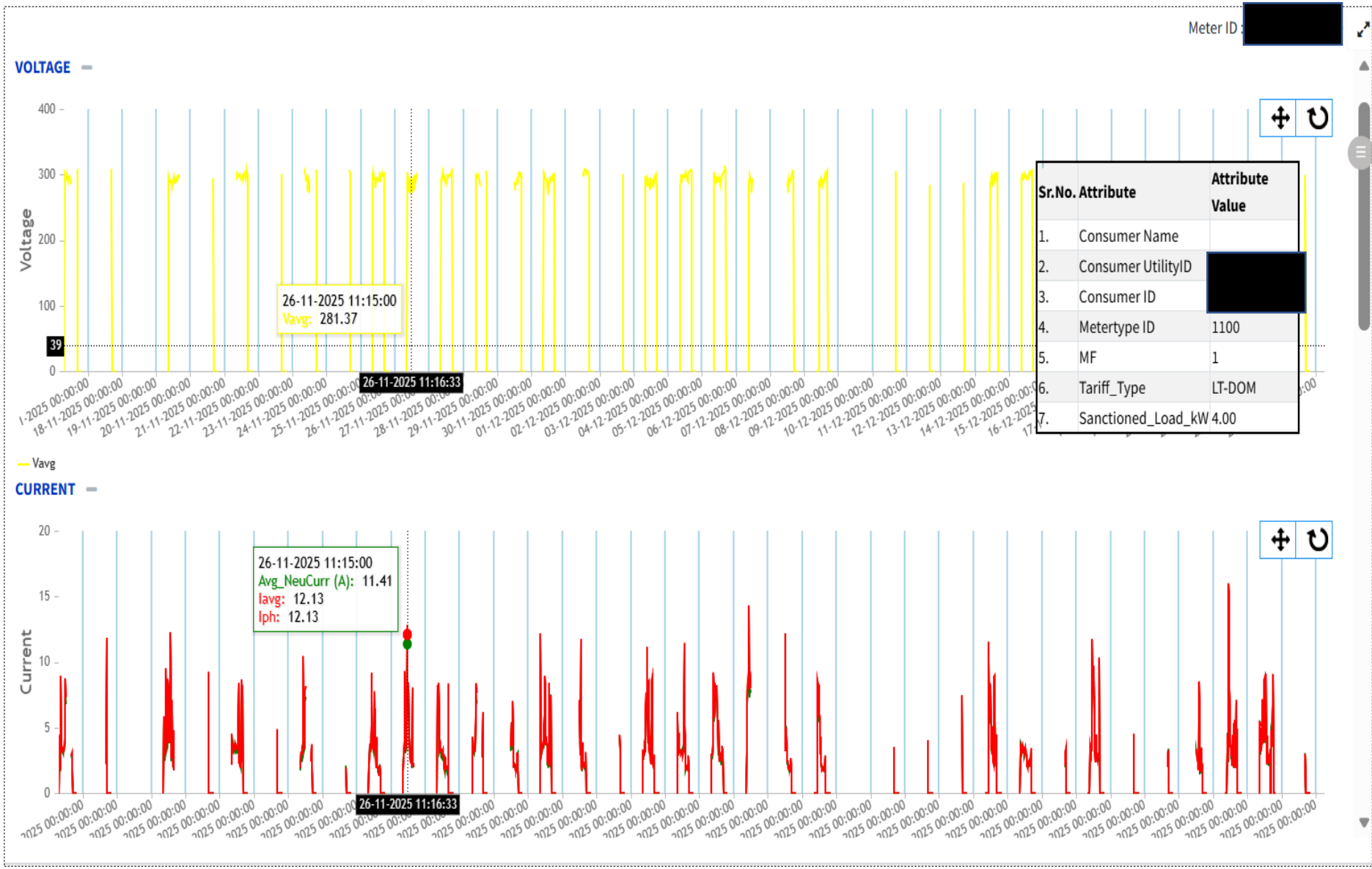
MDAG Remark

- Multiple power failure event logged.
- No data observed in load survey data daily from 5:00 PM to 9:00 AM.
- Daily power supply restore at 9:00 AM till 5:00 PM.
- Load around 10 Amp observed in same period.
- Pattern observed since the meter is installed.

**Site Findings**

**Meter Found Tampered with Light Detect Remote (LDR) Circuit**

**Found 10 cases of such type in a particular area**




| Sr.No. | Attribute | Attribute Value |
|---|---|---|
| 1. | Consumer Name | |
| 2. | Consumer UtilityID | [redacted] |
| 3. | Consumer ID | |
| 4. | Metertype ID | 1100 |
| 5. | MF | 1 |
| 6. | Tariff_Type | LT-DOM |
| 7. | Sanctioned_Load_kW | 4.00 |

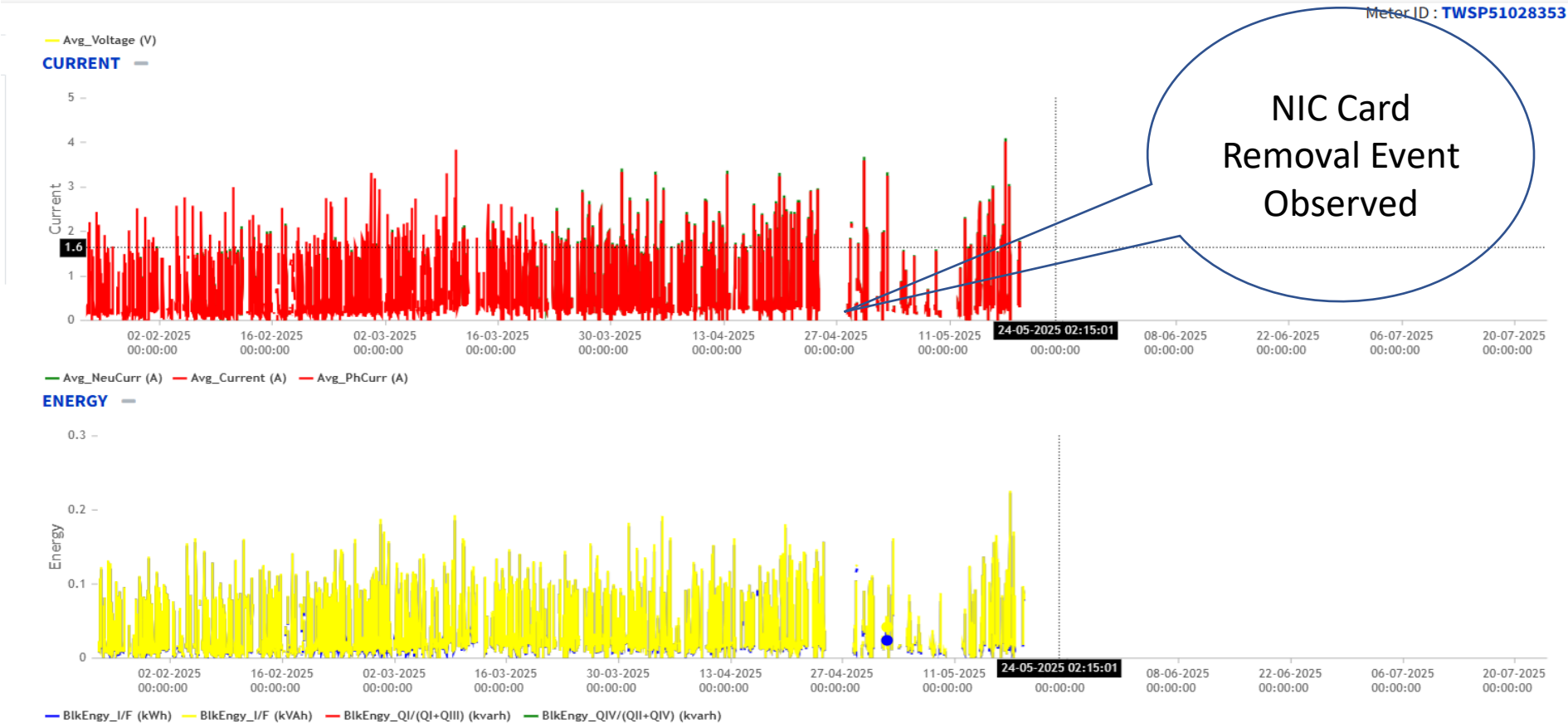
# USE CASE / CASE STUDY

MDAG Remark

After NIC card removal event, there was a sharp consumption Drop

**Site Findings**

**Meter Found Tampered from back side**



# Take Away

| Direct Benefits | MDAG Contribution | UOM |
|---|---|---|
| Auto Analysis of 3 PH & 1 PH Smart Meters | 85 | % |
| Hit Rate of Theft/Anomaly Detection | 52 | % |
| Under Recording (Addition Billing) | 11 | MU |
| Theft(135/126) | 1173 | count |
| Estimated Revenue Generated | 12 | Cr |
| Billing Efficiency Improved | ~1 | % |

## Indirect Benefit

- ✓ Earlier enforcement depended on random checks, leading to low hit rates and high effort.
- ✓ Data-driven MDAG leads now enable targeted, efficient, and faster inspections. Effectiveness of Field teams increases which save time, efforts, and operational cost with precise anomaly detection.
- ✓ Better detection has created **a strong deterrent, improving consumer discipline and reducing meter/asset damage.**

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# *THANK YOU*

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)